



field guide · seeing with light · 6 projects · jun 2026

> build six things that see with light_

Six EE Projects That See With Light

Lasers, photodiodes, and time-of-flight – six builds that measure, map, and talk using light. From a beam-break speed trap to a spinning LiDAR that draws your room as a point cloud, to sending audio across the room on a laser. Cheap optics; real photonics signal processing.

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● By Enes · @1nska · Electrical Engineer – a Build List field guide

Optical Break-Beam Speed Trap

Fire a laser across a gap and clock anything that crosses it – in real units

\$15

beginner

1 weekend

why this matters

The simplest thing you can do with light is notice when it stops arriving. Point a laser at a photodiode and you have a beam; break it and a voltage collapses in microseconds. Put two beams a known distance apart, time how long an object takes to cross from one to the other, and you have a speed trap – clock a Nerf dart, a dropped ball, a swinging door. It is the cheapest possible photonics build, the response is instant and visual, and it teaches the photodetection front end that every later project in this guide depends on.

WHAT THIS PROVES TO A RECRUITER

You understand photodiode and phototransistor detection, turning a light level into a clean logic edge with a comparator, microsecond-scale timing on a microcontroller, and converting a time interval into a physical measurement – the core of optical sensing and instrumentation.

→ the job this maps to: optical sensing & instrumentation – industrial automation, robotics, and test-and-measurement teams.

! LASER EYE SAFETY

Lasers can damage eyes. Use only low-power Class 1 / 2 modules (a few mW), never look into the beam or aim it at anyone, and remember infrared lasers are invisible – you cannot tell they are on. Set beams below eye level and keep this in mind for every build in this guide.

how it works

- ▶ **Light to current:** A photodiode (or phototransistor) turns incident light into a small current. A laser aimed at it holds that current high; break the beam and it collapses. A load resistor turns the current into a voltage you can read.
- ▶ **Clean edge with a comparator:** The raw photodiode voltage is analog and noisy. A comparator (or Schmitt trigger) with a threshold turns “beam present / broken” into a crisp digital edge – immune to ambient drift – that a microcontroller can timestamp.
- ▶ **Two beams, one interval:** Place two beams a measured distance d apart along the path. The object breaks beam A, then beam B; the microcontroller measures Δt between the two edges. Speed = $d / \Delta t$.
- ▶ **Microsecond timing:** For a fast projectile, Δt is sub-millisecond, so you use hardware timers and interrupts (input capture), not slow polling. The resolution of your timer sets the resolution of your speed.
- ▶ **Rejecting ambient light:** Sunlight and room light add a DC offset and flicker. A focused laser well above the ambient level, a threshold set between the two, and a shielded detector keep the beam-break reliable indoors and out.

bill of materials

component	part / spec	qty	~cost
Laser diode module	5 mW red dot laser (×2)	2	\$4
Photodiode / phototransistor	BPW34 or a phototransistor (×2)	2	\$2
Comparator	LM393 (dual)	1	\$1
Microcontroller	Arduino / ESP32	1	\$8
Passives + breadboard	resistors, wiring	—	\$3

build steps

1 Build one beam-detector

Aim a laser at a photodiode across a small gap, with a load resistor to make a voltage. Read it on a scope or ADC and confirm the voltage drops hard when you block the beam.

2 Add a comparator

Feed the photodiode voltage to an LM393 with a threshold set between “beam” and “blocked.” Confirm a clean HIGH / LOW edge as you break the beam, with no chatter.

3 Duplicate for two gates

Build a second identical beam-detector and mount both a known distance d apart along the path, lasers aligned onto their detectors. Measure d precisely – it sets your accuracy.

4 Time the crossing

On the microcontroller, capture the timestamp of beam-A break and beam-B break with hardware timing, compute Δt , and print it. Drop a ball through both beams to test.

5 Convert and display

Compute $\text{speed} = d / \Delta t$ in m/s and mph, show it on serial or an OLED, and add an “armed” state. Now clock darts, balls, or anything that crosses – your optical speed trap.

interview questions

- How does a photodiode convert light into an electrical signal?
- Why use a comparator instead of reading the photodiode voltage directly?
- How do two beams a known distance apart yield a speed measurement?
- What sets the speed resolution of your timing, and how do you improve it?
- How do you keep a beam-break sensor reliable under changing ambient light?
- What is the difference between a photodiode and a phototransistor for this job?

Laser Rangefinder (Time-of-Flight LiDAR)

Measure distance with a pulse of light – the sensor inside every self-driving car

\$25

intermediate

1 weekend

why this matters

Radar measures distance with radio; LiDAR does it with light. A time-of-flight sensor fires a pulse of infrared light, measures how long it takes to bounce back, and multiplies by the speed of light to get range – millimetre-accurate, dozens of times a second. Building one around a module like the VL53L1X gets you a working laser rangefinder in an afternoon, and understanding what it does underneath – pulse timing versus phase, the speed of light, photon statistics – is the foundation of every autonomous-vehicle and robotics perception stack. This is the headline build, and it feeds straight into the scanner in project 3.

WHAT THIS PROVES TO A RECRUITER

You understand time-of-flight ranging, the difference between direct (pulsed) and indirect (phase) ToF, why timing light is hard, signal-to-noise and averaging, and the I2C sensor integration real perception systems are built on. LiDAR is one of the hottest hiring areas in EE right now.

→ **the job this maps to:** autonomous vehicles, robotics, and LiDAR engineering – AV companies, robotics firms, and LiDAR sensor makers (a direct companion to radar roles).

how it works

- ▶ **Time-of-flight, literally:** The sensor emits a short IR pulse and measures the round-trip time t to the target. $\text{Distance} = c \cdot t / 2$. Light travels ~ 30 cm per nanosecond, so the timing must be picosecond-class – which is why it lives inside a dedicated chip, not your Arduino.
- ▶ **Direct vs indirect ToF:** Direct ToF times a sharp pulse's return (good at long range). Indirect (AMCW) ToF modulates the beam and measures the phase shift of the returning light to infer time – easier electronically, and what many compact modules and depth cameras use.
- ▶ **The SPAD detector:** Modules like the VL53L1X use single-photon avalanche diodes that register individual photons, plus on-chip timing, to resolve those tiny intervals and reject ambient light by histogramming returns.
- ▶ **Range, accuracy, averaging:** A single shot is noisy, so the sensor fires many pulses and histograms them to pull the true range out of photon noise. More averaging means better accuracy but a lower update rate – the classic ranging tradeoff.
- ▶ **Reading it over I2C:** The module does the physics and hands you a distance in millimetres over I2C. Your job is configuration (timing budget, region of interest, range mode) and using the result – exactly the integration work real perception engineers do.

BUILD NOTE

Don't try to time light yourself – picosecond timing needs a dedicated chip. Start with a VL53L1X (short range, indoor) or a TFmini (longer range); both hand you millimetres over I2C, so you focus on using the data and, in project 3, scanning it.

key design decisions

- **Why light instead of ultrasound for ranging?** Light gives a tight beam (fine angular resolution), works at long range, and is fast – you can scan it into a detailed map. Ultrasound is cheaper but has a wide beam and low resolution. That tight beam is precisely what makes the LiDAR mapping in project 3 possible.
- **Why does timing light need a dedicated chip?** Light covers 30 cm per nanosecond, so resolving centimetres means timing to tens of picoseconds – far beyond a microcontroller. SPAD arrays and on-chip time-to-digital converters do this; you buy that capability in the module rather than building it.
- **Why average so many pulses?** Each returned pulse is buried in ambient light and shot noise. By firing thousands of pulses and histogramming arrival times, the sensor finds the true peak and rejects background – trading update rate for accuracy, which you tune with the timing budget.

bill of materials

component	part / spec	qty	~cost
ToF LiDAR module	VL53L1X (~4 m) or TFmini (~12 m)	1	\$12
Microcontroller	ESP32 / Arduino (I2C)	1	\$8
Display	OLED for live distance	1	\$4
Passives + breadboard	wiring	—	\$2

build steps

1 Wire the ToF module

Connect the module over I2C (SDA / SCL + power), load the vendor library, read a single distance, and confirm it tracks your hand moving toward and away from the sensor.

2 Tune the timing budget

Set the range mode and timing budget (short / medium / long). Watch how a longer budget improves accuracy and stability but slows the update rate — feel the accuracy-vs-speed tradeoff directly.

3 Calibrate against a tape

Compare readings to a real tape measure at several distances. Note where it's accurate, where it drifts, and how target colour and reflectivity change the reading. Apply any offset.

4 Live distance readout

Show the distance on an OLED with a fast refresh, plus min / max hold. Point it around the room and watch it measure walls, objects, and the ceiling.

5 Stress-test the limits

Find its max range on a dark vs white target, test it outdoors (ambient IR hurts range), and note the failure modes. You now understand a ToF sensor well enough to scan with it next.

interview questions

- Write the relationship between round-trip time and distance for a ToF rangefinder.
- What is the difference between direct (pulsed) and indirect (phase) time-of-flight?
- Why does ranging with light require picosecond-class timing, and how is it achieved?
- What is a SPAD, and why is it useful for a ToF sensor?
- Why does a ToF sensor average many pulses, and what does that trade off?
- How do target reflectivity and ambient light affect a LiDAR range reading?

Spinning LiDAR Room Scanner

Spin a rangefinder and watch your room appear as a point cloud

\$45

intermediate

1-2 weekends

why this matters

One distance reading is a number; thousands of them swept through a circle is a map. Mount your time-of-flight sensor on a rotating platform, log distance against angle as it spins, and plot the result – the outline of your room appears as a 2D point cloud, exactly like the LiDAR on a robot vacuum or the spinning sensor on an early self-driving car. This is the most visual build in the guide and the bridge from “I measured a distance” to “I perceived a space,” which is the heart of robot navigation and SLAM.

WHAT THIS PROVES TO A RECRUITER

You understand synchronising a sensor reading to an angular position, polar-to-Cartesian conversion, point-cloud construction, and the data path from a raw sensor to a usable map – foundational robotics-perception and SLAM groundwork.

→ **the job this maps to:** robotics, autonomous navigation, and perception engineering – robot-vacuum and AMR makers, drone companies, and SLAM teams.

how it works

- ▶ **Distance plus angle:** A LiDAR map is just many (angle, distance) pairs. Rotate the ToF sensor and, at each angle, take a reading. The accuracy of your angle (from the motor) and distance (from the sensor) sets the quality of the map.
- ▶ **Knowing the angle:** A stepper gives you angle for free (count steps); a DC motor needs an encoder or a once-per-revolution index mark to know where “zero” is. Synchronising each distance sample to the correct angle is the core challenge.
- ▶ **Getting data off the spinner:** Either run a slip ring for power and data, or – simpler – put the microcontroller and sensor on the rotating platform and stream readings out over Bluetooth or Wi-Fi, with only power crossing the slip ring (or a battery onboard).
- ▶ **Polar to Cartesian:** Each (θ, r) becomes a point $x = r \cdot \cos\theta$, $y = r \cdot \sin\theta$. Plotting all the points for one revolution draws the walls and objects around the sensor – your 2D point cloud.
- ▶ **From scan to map:** Stack revolutions and you can filter noise, detect walls, and (with motion) start stitching scans together – the first step toward SLAM, where a robot builds a map while tracking its own position.

BUILD NOTE

Don’t machine a precision scanner. A stepper (or geared DC motor with an encoder) plus your project-2 ToF module is enough; send data off the spinning platform over a slip ring, or mount the electronics on the rotor and stream over BLE / Wi-Fi.

bill of materials

component	part / spec	qty	~cost
Project 2 ToF module	VL53L1X / TFmini	1	\$12
Rotating platform	stepper + driver, or geared motor + encoder	1	\$15
Slip ring (optional)	power / data across the rotor	1	\$8
Microcontroller	ESP32 (BLE / Wi-Fi streaming)	1	\$8
Mount + base	3D-printed or laser-cut	1	\$2

build steps

1 Spin the sensor

Mount the ToF module on the stepper / motor platform and get it rotating smoothly at a steady rate. Confirm the sensor still reads distance while spinning.

2 Tag readings with angle

Sync each distance sample to the motor angle (step count, or encoder + index pulse). Verify that a known object in a known direction shows up at the right angle.

3 Stream the data out

Send (angle, distance) pairs to a laptop over BLE / Wi-Fi (or through a slip ring). Confirm a continuous stream of pairs for each full revolution.

4 Plot the point cloud

On the laptop, convert each pair to x, y and plot one revolution. Your room's outline should appear. Tune speed and sample rate for a clean scan.

5 Refine and map

Average multiple revolutions to cut noise, mask the mount's own structure, and watch the map sharpen. Move the scanner and see how the cloud shifts – the doorway to SLAM.

interview questions

- What two measurements combine to make a single LiDAR map point?
- How do you determine the angular position of each distance reading?
- What are the options for getting data off a rotating sensor, and their tradeoffs?
- Write the polar-to-Cartesian conversion used to plot a scan.
- What is SLAM, and how does a spinning rangefinder feed into it?
- How does rotation speed trade off against map density and quality?

Laser Data Link (Free-Space Optical Comms)

Send audio and data across the room on a beam of light

\$20

intermediate

1 weekend

why this matters

Fibre-optic internet is just light carrying data down a glass thread – and you can do the same thing through open air with a laser and a photodiode. Modulate a laser's brightness with an audio or digital signal, point it at a photodiode across the room, and recover the signal on the far side: a free-space optical link, the same idea as Li-Fi and satellite laser comms. It is a hands-on lesson in modulation, analog front ends, and why light is such a high-bandwidth carrier – and unlike radio it needs no licence and barely leaks past the beam.

WHAT THIS PROVES TO A RECRUITER

You understand intensity modulation of an optical source, transimpedance photodiode receivers, bandwidth and signal-to-noise in an analog link, and the basics of free-space optics – which maps onto optical-comms, fibre, and photonics engineering.

→ **the job this maps to:** optical communications and photonics engineering – fibre and datacom companies, free-space-optics and satellite-comms teams.

how it works

- ▶ **Intensity modulation:** The simplest optical link varies the laser's brightness in step with your signal. For audio, you add the audio to the laser's drive current so the light flickers with the waveform; a photodiode on the far end sees that flicker.
- ▶ **The transimpedance receiver:** The photodiode produces a tiny current proportional to received light. A transimpedance amplifier (op-amp + feedback resistor) converts it to a usable voltage with enough bandwidth for your signal – the heart of every optical receiver.
- ▶ **Analog vs digital:** Sending audio (analog AM on light) is the easy first version. Sending digital data (on-off keying, UART over light) is the next step: now you care about bit rate, thresholding, and clean edges rather than fidelity.
- ▶ **Bandwidth, and why light wins:** The optical carrier is hundreds of terahertz, so the available bandwidth dwarfs radio – which is why fibre and laser links carry so much data. Your build's actual speed is limited by the emitter and receiver electronics, not the light.
- ▶ **Alignment and noise:** A tight beam makes the link private and interference-resistant, but you must aim it. Ambient light adds noise and offset, so you AC-couple the receiver (and modulate above the flicker of room lights) to keep the signal clean.

bill of materials

component	part / spec	qty	~cost
Laser or IR LED	5 mW laser module or high-speed IR LED	1	\$4
Photodiode	BPW34 or similar, fast	1	\$2
Op-amp	for the transimpedance amplifier	1	\$1
Driver / audio parts	transistor driver, audio source	—	\$5
Microcontroller (digital)	optional, for UART-over-light	1	\$8

build steps

1 Build the transmitter

Drive the laser / LED with a transistor so its brightness follows your signal. For audio, sum an audio source into the drive current; confirm the light measurably flickers with the waveform.

2 Build the receiver

Wire a photodiode into an op-amp transimpedance amplifier. Point a steady light at it and confirm the output tracks brightness; check the bandwidth is enough for your signal.

3 Close the link with audio

Aim the transmitter at the receiver across a gap, feed music into the transmitter, and amplify the receiver output to a speaker. You should hear the audio carried on the beam.

4 Reject ambient light

AC-couple the receiver and shield it so room light doesn't swamp the signal. Note how alignment and distance change the received level.

5 Go digital (optional)

Switch to on-off keying and send UART bytes over the beam between two microcontrollers. Measure the maximum reliable bit rate and see what limits it.

interview questions

- What is intensity modulation, and why is it the simplest optical-link scheme?
- Why is a transimpedance amplifier the standard photodiode receiver front end?
- Why does an optical carrier offer so much more bandwidth than radio?
- What limits the actual data rate of a simple free-space optical link?
- How do you keep an optical receiver from being swamped by ambient light?
- What are the advantages and drawbacks of a tight optical beam for comms?

Laser Vibrometer

Read a surface's vibration with a reflected beam – non-contact sensing with light

\$30

advanced

1–2 weekends

why this matters

Point a laser at a reflective surface, catch the reflected dot on a photodiode, and any vibration of that surface changes the light reaching your detector – letting you read the surface's motion without touching it. This is laser vibrometry, a real engineering technique for measuring how speakers, motors, and structures vibrate without loading them down. Building one teaches optical detection of tiny displacements, signal recovery from a noisy photodetector, and the limits of non-contact sensing. It is one of the most striking demonstrations of how much information a single beam of light carries.

WHAT THIS PROVES TO A RECRUITER

You understand optical displacement and vibration sensing, recovering a small AC signal from a photodetector, filtering and amplification, and the practical limits of non-contact measurement – which maps onto optical metrology, acoustics, and structural-health-monitoring work.

→ **the job this maps to:** optical metrology and sensing engineering – acoustics labs, structural-health monitoring, and precision-measurement teams.

! MEASURE YOUR OWN SURFACES ONLY

This is a vibration-sensing technique for surfaces and devices you own – a speaker cone, a motor housing, a panel on your own bench. Do not point it at other people's windows, rooms, or property to capture sound or conversations: that is eavesdropping, it is illegal in most places, and it is not what this build is for. Keep it on your own equipment.

how it works

- ▶ **Light carries surface motion:** When a laser reflects off a surface, the reflected beam's direction and the light coupled back into your detector depend on the surface's exact position. Vibrate the surface and the photodiode sees a matching AC signal riding on the reflection.
- ▶ **The detector geometry:** A simple setup catches the reflected dot at the edge of a photodiode (or through a slit) so small angular shifts change how much light lands on the sensor – converting tiny surface motions into a measurable voltage swing.
- ▶ **Recovering a tiny signal:** The vibration signal is small and sits on a large DC level from the steady reflection. AC-coupling, a low-noise amplifier, and band-pass filtering pull the vibration (an audio-band signal from a speaker, say) out of the background.
- ▶ **Why it is hard:** The signal depends heavily on surface reflectivity, alignment, and ambient light, and it is easily swamped by noise or a poor reflector. Stable mounting and a cooperative, shiny target are what make it work.
- ▶ **What the pros do:** True laser Doppler vibrometers interfere the reflected beam with a reference to read sub-wavelength motion precisely. Your build is the intuition-level version of that instrument – same idea, far simpler optics.

bill of materials

component	part / spec	qty	~cost
Laser module	5 mW red dot laser	1	\$3
Photodiode	BPW34 or similar	1	\$2
Low-noise amp	op-amp + band-pass stage	1	\$3
Capture	audio interface or ESP32 ADC	1	\$8
Mounts / optics	rigid stands, small mirror, your own test surface	–	\$6

build steps

1 Set up a stable reflection

Mount the laser and a target surface (e.g. a small mirror taped to a speaker you own) rigidly, and catch the reflected dot on the photodiode. Stability matters more than anything else here.

2 Build the receiver

Convert the photodiode current to voltage, AC-couple it, and add a low-noise band-pass amplifier for the band you care about (audio for a speaker). Confirm a quiet baseline when the surface is still.

3 Excite the surface

Play a tone through your speaker and watch the photodiode signal pick up a matching AC waveform. Sweep the tone and see the response track it.

4 Recover the signal

Filter and amplify so you can plot or listen to the recovered vibration. Compare it to the original tone to see how faithfully light captured the motion.

5 Explore the limits (responsibly)

On your own equipment, test how reflectivity, distance, and alignment change the result, and where it breaks down. Keep it pointed only at surfaces you own.

interview questions

- How can a reflected laser beam encode a surface's vibration?
- Why is the recovered vibration signal small, and how do you extract it?
- What role does detector geometry play in sensing tiny surface motions?
- Why do reflectivity and alignment dominate this measurement's quality?
- How does a real laser Doppler vibrometer achieve sub-wavelength precision?
- What are the legitimate engineering uses of non-contact vibration sensing?

Structured-Light 3D Scanner

Sweep a laser line across an object and rebuild it in 3D

\$40

advanced

2 weekends

why this matters

A camera sees a flat image; add a laser and geometry and you can recover depth. Project a thin laser line onto an object, watch from a camera offset to one side, and the line bends along the object's contours – measure that bend and triangulation gives you the surface's 3D shape. Sweep the line across the object and you reconstruct a full 3D model, the same structured-light principle behind many 3D scanners and depth cameras. It is the most ambitious optical build here and a direct lesson in the camera-plus-light geometry at the core of 3D vision.

WHAT THIS PROVES TO A RECRUITER

You understand triangulation, camera-laser calibration, extracting a laser line from an image, and reconstructing 3D points from 2D observations – core 3D-vision, metrology, and depth-sensing engineering.

→ **the job this maps to:** 3D vision, metrology, and depth-sensing engineering – 3D-scanning and CV companies, robotics perception, and industrial-inspection teams.

how it works

- ▶ **Triangulation, not timing:** Unlike ToF, structured light gets depth from geometry. The laser, the camera, and a point on the object form a triangle; knowing the laser-to-camera baseline and angles, you solve for the point's distance – the same principle as your two eyes judging depth.
- ▶ **The bending line:** A flat reference plane shows the laser line straight. An object pushes parts of the line sideways in the camera image; how far each pixel of the line shifts encodes how far that part of the surface is from the reference – a depth profile in one shot.
- ▶ **Finding the line in the image:** For each camera row you find the brightest (reddest) pixel – the laser line's position, to sub-pixel accuracy. Robust line extraction under ambient light is the key image-processing step.
- ▶ **Calibration:** You must know the geometry: the angle of the laser plane and the camera's parameters and position. A calibration step using known reference distances turns pixel shifts into real millimetres.
- ▶ **Sweeping to a model:** One line gives one slice. Rotate the object on a turntable (or sweep the line) while capturing slices, then stack them into a full 3D point cloud or mesh you can view and export.

BUILD NOTE

Don't build custom optics. A line-laser module (a dot laser with a cylindrical lens), a webcam offset to one side, and a turntable or slow manual sweep are enough; the geometry and the image processing are where the real work – and learning – live.

bill of materials

component	part / spec	qty	~cost
Line laser module	dot laser + cylindrical lens (line generator)	1	\$6
Camera	USB webcam	1	\$12
Turntable	stepper-driven platform (optional)	1	\$15
Frame / mount	rigid laser + camera baseline	1	\$5
Laptop	OpenCV for line extraction + reconstruction	1	–

build steps

1 Build the rig

Fix a line laser and a webcam a known distance apart (a baseline), with the camera viewing the laser line on a reference plane. Rigidity is essential – any flex corrupts the geometry.

2 Extract the laser line

In OpenCV, isolate the laser line (colour threshold) and find its position per image row to sub-pixel accuracy. Confirm a clean line trace even with some room light.

3 Calibrate the geometry

Using known distances on the reference plane, work out how a pixel shift of the line maps to a real depth. Validate against an object of known size.

4 Capture one profile

Place an object so the line drapes over it; the bent line is one depth slice. Convert the bend to 3D points and plot that single profile.

5 Sweep and reconstruct

Rotate the object on a turntable (or sweep the line) capturing many slices, stack them into a point cloud, and view / export your 3D model. Refine lighting and calibration for a clean scan.

interview questions

- How does structured-light scanning recover depth without timing light?
- Why does a laser line appear to bend over a 3D object in the camera view?
- How do you extract a laser line from an image to sub-pixel accuracy?
- Why is camera-laser calibration essential, and what does it determine?
- How do single depth profiles combine into a full 3D model?
- What are the tradeoffs between structured light, ToF, and stereo for depth?

> wrap

Light is the fastest instrument you own – point it and read what comes back.

A break-beam, a time-of-flight pulse, a spinning rangefinder, a modulated beam, a reflected dot, a bent laser line – six ways to turn light into information: distance, shape, motion, and data. The optics are cheap (a \$3 laser, a \$2 photodiode, a webcam), and just like radar, the real skill is in the front end and the signal processing: the detection, the timing, the triangulation, the filtering. That is exactly what optical-sensing, LiDAR, and photonics teams hire for.

// the skill ladder

Build them in order and each one hands you the technique the next one needs:

photodetection → time-of-flight → LiDAR scanning → optical comms → vibrometry → structured light

! LASER SAFETY, EVERY TIME

Use only low-power Class 1 / 2 modules, never look into a beam or aim one at anyone, and respect that infrared lasers are invisible. And keep the vibrometer pointed only at surfaces you own – not anyone else's windows.

DO THIS

Build even one of these. Capture the output – a speed readout, a point cloud of your room, audio carried on a beam, a 3D scan – and put it on GitHub with a clear README, the geometry or calibration you used, and an honest note on scope. One optics demo and one clean repo prove you can pull real information out of light.

Full build notes, parts links, and the next field guide go out in **The Build List**. The people actually shipping these – and trading alignment and calibration tricks – are in **Ohm's Club**.

– Enes (@1nska) · Electrical Engineer